

length of 3 months and an operative length of 6 months, the tick size for May 2018 through August 2018 is set by the stock's TWAQS from January 2018 through March 2018. The statistics in the table are determined by stocks' trading during the May 2018 through August 2018 period (the operative period). This process is repeated on a rolling basis for every month from January 2018 to June 2022, and the table summarizes results across all months.

TWAQS is determined by computing the time-weighted quoted spread during regular trading hours as computed by the WRDS intra-day indicators for every sym_root and sym_suffix combination in the dataset. When calculating a stock's TWAQS during an evaluation period, the stock's daily TWAQS is averaged across all trading days in the evaluation period.

When assigning volume to a TWAQS bucket in an operative period, the TWAQS on a given day for a particular stock is used. That is, if a stock trades with a TWAQS of \$0.011 on Monday but the same stock has a TWAQS of \$0.016 on Tuesday, then its volume on Monday is assigned to the sub-\$0.015 category while its Tuesday volume is assigned to the over-\$0.015 category in the operative period.

The universe of securities in the WRDS intra-day indicators dataset is used.

Panel A computes the fraction of total aggregate share volume that occurs in stocks that would have been assigned a \$0.01 tick yet subsequently trade at a TWAQS of under \$0.015 in the operative period. These stocks would benefit from a \$0.005 tick instead of a \$0.01 tick.

Panel B computes the fraction of total aggregate share volume that occurs in stocks that would have been assigned a \$0.005 tick yet subsequently trade at a TWAQS of over \$0.015 in the operative period. These stocks may not benefit from the \$0.005 tick.

Panel C computes false negative rates. The false negative rate is the fraction of share volume that is assigned a tick of \$0.01 among the share volume that trades with a TWAQS under \$0.015 in the operative period.

Panel D computes the false positive rates. The false positive rate is the fraction of share volume that is assigned a tick of \$0.005 among the share volume that trades with a TWAQS above \$0.015 in the operative period.

Table 11 further explores commenters' discussions about the risk of placing too much emphasis on transient market conditions. In particular, Panels A and B of Table 11 computes rates of false negatives and positives—similar to panels C and D of Table 10—but does so using only the evaluation period that ends with March of 2020. The one-month evaluation period includes only March of 2020; the three-month evaluation period covers January to March of 2020; the five-month evaluation period covers November of 2019 to March of 2020; the twelve-

month evaluation period covers April of 2019 to March of 2020. In this way, Panels A and B of Table 11 demonstrate what may happen if an unusual market event causes an inappropriate tick assignment. These panels show that a one-month evaluation period performs particularly poorly when that month is March of 2020. Specifically, the one-month evaluation period exhibits a substantially higher false negative rate.¹³⁷⁸ A three-month evaluation period consistently has the lowest false negative rate—it has the benefit of recent data without an over-reliance on short-term fluctuations. Finally, the false positive rates in Panel B approximately double when moving from a six-month operative period to a twelve-month operative period, indicating that infrequent tick updating can lead to stocks getting stuck with an inappropriately low tick for an extended period.

Panels A and B of Table 11 add to our understanding of the tradeoff presented by the evaluation period. Table 10 suggests that short evaluation periods tend to reduce false negatives but increase false positives. Given that false negatives (vs. false positives) tend to be more prevalent and vary more across evaluation periods, Table 10 suggests that a one-month evaluation period may be best. However, Table 11 shows that a one-month evaluation period may substantially increase false negatives when the evaluation period includes a period of unusual market stress.

While Panels A and B of Table 11 includes March of 2020 in the evaluation period, one commenter also provided, in the context of the Proposing Release, analysis using March of 2020 in the operative period. This commenter stated that during times of market stress, “many symbols would be trading with far more price levels intra-spread than contemplated under the Proposal,

¹³⁷⁸ The false positive rates are generally lower than usual (e.g., below those in Panel B), likely because quoted spreads narrowed after March of 2020.

thereby further increasing the liquidity-related harms...” The commenter further showed that, were the proposed rule in effect, most symbols receiving a tick reduction would have experienced over ten intra-spread ticks during March of 2020, and many stocks would have experienced over twenty intra-spread ticks.¹³⁷⁹

The Commission acknowledges that quoted spreads generally widen during periods of market stress, and this may result in stocks trading with more intra-spread ticks than is desirable; the rise in intra-spread ticks may then compound the market stress. Relative to the proposal, the amendments reduce the severity of such an outcome by reducing the number of stocks that receive a tick reduction, and by reducing the size of the tick reduction. To further examine this issue, the Commission conducts its own analysis with March of 2020 as the operative period. In particular, the Commission simulates tick assignment in February of 2020 under the parameters of the amendments, and then examines the outcome during March of 2020. The simulation is done for a range of evaluation periods—a one-month evaluation period uses January of 2020 to determine which stocks receive the \$0.005 tick in March (allowing for a one-month lag), a three-month evaluation period uses November of 2019 to January 2020 to assign ticks, a five-month evaluation period uses September of 2019 to January 2020, and a twelve-month evaluation period uses February of 2019 to January 2020. The Commission then examines the fraction of aggregate share volume that would have received a \$0.005 tick under the amendments yet traded with a wide quoted spread during an operative period of March 2020.

Results are presented in panel C of Table 11. Each row corresponds to an evaluation length of 1, 3, 5, or 12 months. Each column corresponds to a quoted spread threshold of \$0.015,

¹³⁷⁹ See Citadel Letter I at 6-7.

\$0.02, or \$0.05. The upper-left number—21.4%—indicates that 21.4% of aggregate share volume in March of 2020 would have occurred with a \$0.005 tick and a TWAQS over \$0.015. With a three-month evaluation period, this fraction halves to 10.6%, further reinforcing the conclusion of Panel A that a one-month evaluation period may do particularly poorly when market conditions suddenly worsen. The Commission further reiterates that stocks receiving a \$0.005 tick are unlikely to be harmed when trading at a quoted spread of \$0.015, so it is unlikely that 10.6% of share volume would have been harmed in March of 2020 were the amendments in place. To further explore this issue, the Commission performs similar calculations with wider quoted spread thresholds of \$0.02 and \$0.05. With a three-month evaluation period, the fraction of aggregate share trading that receives a \$0.005 tick and trades at a quoted spread over \$0.02 is 6.4%; this reduction (from 10.6% trading at a quoted spread over \$0.015) indicates that a substantial proportion of the false positive trading during March of 2020 is occurring with 3-4 intra-spread ticks, which is generally in line with commenters’ views on the optimal number of intra-spread ticks.¹³⁸⁰ Finally, with a three-month evaluation period, the fraction of aggregate trading that receives a \$0.005 tick and trades at a quoted spread over \$0.05 is 1.2%. Analysis in the Proposing Release and herein suggests that this 1.2% of volume is at an increased risk of a reduction in market quality due to having over ten intra-spread ticks.¹³⁸¹

Table 11: Effect of the evaluation period on inappropriate tick assignment around March 2020				
Panel A: False negative rates using March of 2020 in the evaluation period				
Length of operative period in months				
	1	3	6	12

¹³⁸⁰ See supra note 1299.

¹³⁸¹ See supra section VII.D.1.b.ii, particularly Figure 2 and surrounding discussion.

Length of evaluation period in months	1	36.2%	35.4%	37.4%	40.1%
	3	21.7%	22.5%	23.7%	23.9%
	5	34.7%	34.9%	36.1%	37.4%
	12	30.0%	31.7%	32.0%	33.9%
Panel B: False positive rates using March of 2020 in the evaluation period					
Length of operative period in months					
		1	3	6	12
Length of evaluation period in months	1	1.7%	2.6%	3.2%	8.0%
	3	3.0%	4.6%	6.0%	12.0%
	5	2.6%	4.1%	5.4%	10.2%
	12	3.6%	5.0%	6.5%	11.0%
Panel C: Fraction of March 2020 share volume with a \$0.005 tick and wide quoted spreads					
Quoted spread threshold					
		AQS > \$0.015	AQS > \$0.02	AQS > \$0.05	
Length of evaluation period in months	1	21.4%	14.4%	3.5%	
	3	10.6%	6.4%	1.2%	
	5	12.4%	7.6%	1.5%	
	12	14.4%	9.2%	2.0%	
^a The Commission simulates the tick assignment procedure under 16 combinations of evaluation and operative period lengths. The methodology and data used for this simulation is described in the note to Table 10.					
In contrast to Table 10, which performed the simulation on a rolling basis for every month from January 2018 to June 2022, this table only simulates tick assignment and outcomes around March of 2020.					
Panel A repeats the false positive calculations of Table 10 as if the tick is assigned in April 2020 (to ensure that March 2020 is in the evaluation period). For example, to compute the statistic for an evaluation length of 3 months and an operative length of 6 months, the tick size for May 2020 through August 2020 is set by the stock’s TWAQS from January 2020 through March 2020. April 2020 is the hypothetical month in which the tick assignment is calculated and acts as the gap between the evaluation period and the operative period. Panel B similarly repeats the false negative calculations of Table 10 as if April 2020 is the month in which the tick assignment is calculated.					

Panel C instead uses March 2020 as the operative period. For example, to compute the statistic for an evaluation length of 3 months, the tick size for March 2020 is assigned by the stock's TWAQS from November 2019 to January 2020, with February of 2020 being the hypothetical month in which the tick assignment is calculated. Panel C computes the fraction of total aggregate share volume that occurs in stocks assigned to a \$0.005 tick, yet trades at a high TWAQS during March of 2020. The fraction is calculated using a range of TWAQS thresholds for trading during March of 2020—the columns correspond to trading with a TWAQS over \$0.015, \$0.02, and \$0.05.

The analysis indicates that a shorter operative period almost always results in fewer errors. By updating the tick more frequently, the tick better reflects changing market conditions—this is shown by increasing rates of both false negatives and positives as the operative period widens.¹³⁸² However, commenters stated that more frequent updates may impose costs in terms of adjustments to algorithms, operations, systems, and an increase in customer complaints. One commenter requested that the Commission “investigate historical rates of change in the TWAS for a variety of trading symbols. This would avoid imposing undue costs on market participants that may be caused by tick size updates that are too frequent but should mitigate the possibility that tick sizes do not update frequently enough and lead to worse trading outcomes.”¹³⁸³

The Commission examines the frequency of tick size updates in Table 12. This table shows the typical number of tick-changes that occur in a 12-month period under each combination of evaluation and operative period length.¹³⁸⁴ Shorter operative periods present a

¹³⁸² In Panels A-D of Table 10 and Panels A-B of Table 11, moving left-to-right along any row results in a monotonic increase in both false positives and false negatives. The only exception to this pattern is in Panel A of Table 11: with an evaluation period of one-month, the false negative rate falls by 0.8% when the operative period increases from one to three months.

¹³⁸³ See Mitre Corp. Letter at 5.

¹³⁸⁴ These estimates of tick changes are likely to be over-estimates because the estimates are constructed from historical data in which stocks were constrained by the \$0.01 tick. Once the adopted tick amendment is implemented, the stocks assigned a \$0.005 tick will be able to trade at quoted spreads below \$0.01; this is

tradeoff, they better tailor the tick size to current conditions for the stock which can improve market quality for the stock, but they impose two costs: first, a shorter operative period implies ticks are updated more frequently during a year, which increases the number of discrete system changes that market participants need to make. Second, a shorter operative period increases the number of tick-changes that stocks experience during a typical twelve-month period—this is seen in Table 12 by the increase in tick changes as one moves right-to-left in any row. More tick changes may result in result in more extensive changes to trading algorithms and may increase customer confusion and complaints. The risk of customer confusion is mitigated by the adoption of a tick size indicator in the regulatory data. This indicator can be directly incorporated into trading algorithms helping to automate the process adjusting trading strategies to different tick sizes for algorithmic traders.¹³⁸⁵

Table 12: Effect of the evaluation period on the median number of tick changes over a 12-month period					
		Length of operative period in months			
		1	3	6	12
Length of evaluation period in months	1	4,722	2,317	1,356	665
	3	2,632	1,969	1,253	653
	5	1,724	1,435	1,153	622
	12	772	680	591	488
^a The Commission simulates the tick assignment procedure under 16 combinations of evaluation and operative period lengths. The simulation is performed on a rolling basis for every month from January 2018 to June 2022. The methodology and data used for this simulation is described in the					

expected to lower the stocks' TWAQS, thereby reducing the probability that a low-tick stock will experience a tick change by crossing over the \$0.015 TWAQS threshold. It is possible, however, that the amended rule's reduction in the access fee cap may cause quoted spreads to widen, though transaction costs are not expected to go up, and thus shift the distribution of quoted spreads toward the \$0.015 threshold; if more stocks trade near the \$0.015 threshold, then there may be more switching as stocks move across the threshold more frequently.

¹³⁸⁵ See *infra* section VII.D.5 for additional discussion of the expected costs associated with the amendments to Rule 612.

note to Table 10. This table computes the number of tick changes that occur over a median 12-month horizon for each combination of evaluation and operative period length.
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To summarize the results of this subsection, Tables 10, 11, and 12 illustrate the tradeoffs inherent in the choice of evaluation and operative periods. With respect to the evaluation period, Table 10 implies that shorter period lengths reduce false negatives but increase false positives; the higher prevalence of false negatives tilts the scale toward a short evaluation period. Table 11, though, shows that a one-month evaluation period may unduly increase the influence of aberrant events, while a three-month evaluation period continues to perform well in unusual market conditions.

With respect to the operative period, Tables 10 and 11 imply that shorter period lengths reduce both false negatives and positives. Table 11 shows a particularly large reduction in false positives when using a six-month operative period instead of a twelve-month period. Focusing on an evaluation period of three-months, Tables 10 and 11 show similar error rates with operative period of three and six months. However, a three-month operative period requires market participants to update their systems twice as often as a six-month period, and Table 12 shows that the number of annual tick changes increases 57% when moving from the six-month to the three-month period.¹³⁸⁶ The amendments, which adopt a three-month Evaluation Period and a six-month operative period, reflect these considerations.¹³⁸⁷

Finally, the Commission examined the effect of a one-month lag between the end of the evaluation period and the subsequent tick assignment. All the results in Tables 10, 11, and 12

¹³⁸⁶ With a three-month evaluation period, a six-month operative period results in 1,253 annual tick changes, while a three-month operative period results in 1,969. The increase is therefore calculated as: $(1969 - 1253)/1253 = 57\%$.

¹³⁸⁷ See 17 CFR 242.612(a)(1), (b)(1) for rule text relating to these periods.

include this one-month lag; the Commission separately calculated Table 10 statistics without the lag. If the lag is removed, then the error rates in Panels A and B of Table 10 fall by a small amount across all combinations of evaluation and operative periods. The lag effectively makes the evaluation period less informative about the operative period, and this effect is stronger when updates are made every month. With respect to magnitudes, removing the lag causes the fraction of trading that is a false negative to decrease by an average of 0.9% with a maximum decrease of 1.2% for the top left cell of Panel A; the false positives in Panel B likewise drop by an average of 0.2% when the lag is removed. On the other side of the scale, the lag may reduce operational burdens on market participants by allowing them time to update their systems before a new tick assignment occurs, and may likewise provide brokers time to give advance notice to customers about which stocks will experience tick changes.¹³⁸⁸ The amendments, which provide a one-month lag, reflect a conclusion that the cost of increased error rates due to the one-month lag is small relative to the benefits that the lag provides for industry testing and adjustment.

e. Additional Effects of a Half-Penny Tick

Some commenters suggested, in the context of the proposal, that a tiered tick size could create confusion in the markets particularly for retail traders.¹³⁸⁹ These commenters stated that retail traders may be at a relative disadvantage because they may get confused by the sub-penny increments. One commenter presented data suggesting that retail traders tend not to use sub-penny non-marketable limit orders even when they can, for instance for orders priced below

¹³⁸⁸ The adopted rule additionally syncs the dates of tick assignment with the dates of new round lot assignments. This further reduces operational burdens on market participants. See discussion in section VII.D.5.a.

¹³⁸⁹ Fidelity Letter at 3; Tastytrade Letter at 18-19.

\$1.00, and that fill rates for such trades tend to go down for retail traders.¹³⁹⁰ The commenter ascribes these findings to retail investors getting confused and being unfamiliar with sub-penny trading increments leading to a competitive disadvantage for these traders with respect to more sophisticated traders that are more familiar with sub-penny increments and thus retail traders using non-marketable limit orders would be more likely to be undercut with a smaller tick.¹³⁹¹ Another commenter suggested that retail investor confusion would lead retail brokers to need to hire additional staff to manage customer confusion and education concerning multiple tick sizes, or to handle phone trades by retail customers confused by the markets.¹³⁹²

The Commission acknowledges some potential for confusion but does not expect the amendments to disadvantage retail traders. First, the comments are in response to the proposal; the adopted amendments have fewer minimum quoting increments than the proposed amendments; indeed, there are only two as opposed to four. Second, any confusion or disadvantage must be evaluated relative to the baseline. Currently the penny tick creates a price floor, which, especially for tick-constrained stocks, advantages faster and generally more sophisticated traders. It also creates incentives for more complex strategies such as those involving alternative venues, again, contributing to complexity and putting less sophisticated investors at a disadvantage. Third, the one-month period of time between measuring the TWAQS and the assigned tick size becoming operational will allow time for broker-dealers to educate

¹³⁹⁰ See Fidelity Letter at 3.

¹³⁹¹ See also O'Brien Letter at 4 and Tastytrade Letter at 20 making similar comments.

¹³⁹² See Tastytrade Letter at 18-19.